Deep Learning for Exoplanet Atmospheric Retrieval: Scaling, Architecture Ablation, and Statistical Validation on the INARA ATMOS Dataset

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Abstract

Retrieving the chemical composition of exoplanet atmospheres from climate model output is a central task for missions such as the James Webb Space Telescope (JWST) and the planned Habitable Worlds Observatory (HWO). Classical Bayesian samplers are accurate but require hours per planet, making them impractical at survey scale. Zorzan et al. (2025) introduced the INARA ATMOS dataset of 3.1 million synthetic rocky-planet atmosphere simulations and demonstrated a proof-of-concept machine learning retrieval. We build on this dataset to provide a systematic empirical study of three model classes—a compact 1D convolutional neural network (CNN), a flat multi-layer perceptron (MLP), and a Random Forest (RF) baseline—across training set sizes from 1,000 to 86,800 samples, with statistical validation across five random seeds.

Our main findings are: (1) the 1D CNN achieves mean $R^2 = 0.9993 \pm < 0.0001$ on ten atmospheric molecules at 86,800 training samples, compared to 0.8295 ± 0.0053 for the RF baseline—a gap that is statistically decisive and reproducible; (2) a flat MLP achieves nearly identical accuracy to the CNN ($\Delta R^2 = +0.0007$) while requiring 2.6× more parameters (963K vs. 368K), establishing that the 1D convolutional inductive bias provides no meaningful advantage over fully-connected layers for CLIMA profiles; (3) both deep models saturate near 50,000 training samples; and (4) H_2O—previously reported as a near-constant, effectively unlearnable molecule at 25,000 samples—achieves $R^2 = 0.999 \pm 0.001$ at 86,800 samples, demonstrating that its apparent degeneracy was a data-scarcity artifact rather than a fundamental limitation.

1 Introduction

Atmospheric retrieval—inferring the chemical composition of a planetary atmosphere from observations or model output—is fundamental to characterising exoplanets as habitable or otherwise. The standard pipeline applies Bayesian sampling (nested sampling, MCMC) to a forward climate model; this is principled but demands hours of compute per planet (Madhusudhan, 2018). Upcoming surveys with JWST (Gardner et al., 2023) and the future Habitable Worlds Observatory will observe thousands of planets, making per-planet Bayesian retrieval intractable.

Machine learning has been proposed as a path to millisecond-scale retrieval. Márquez-Neila et al. (2018) demonstrated that a Random Forest trained on synthetic spectra can retrieve abundances orders of magnitude faster than Bayesian methods. More recent work applies normalising flows (Vasist et al., 2023) and flow matching (Gebhard et al., 2024) to recover full posterior distributions over atmospheric parameters. These methods are powerful but complex; their accuracy on climate-model profiles at varying data scales has not been systematically characterised.

Zorzan et al. (2025) released INARA ATMOS, a dataset of 3.1 million synthetic rocky-planet CLIMA profiles (Arney et al., 2016) paired with ground-truth $\log_{10}$ surface mixing ratios for twelve molecular species,

specifically designed to benchmark ML retrieval. Their proof-of-concept showed that a simple model can outperform Bayesian sampling in speed. What remained open were: (i) how deep-learning accuracy scales with training data, (ii) whether spatial structure in CLIMA profiles (12 channels $\times$ 101 altitude levels) genuinely benefits from a convolutional architecture, and (iii) whether all twelve molecules are learnable with sufficient data.

We address all three questions with controlled experiments on a 124,000-sample subset of INARA ATMOS. Our contributions are:

1. A compact 1D CNN achieving $R^2 > 0.999$ on ten molecules at 86,800 training samples (Section 6).
2. A controlled architecture ablation (CNN vs. MLP with identical training recipe) showing that convolution adds negligible accuracy benefit ($\Delta R^2 = 0.0007$) for CLIMA profiles (Section 6.4).
3. A scaling analysis over six training set sizes, showing performance saturation near 50,000 samples for both deep models (Section 6.2).
4. Statistical validation across five random seeds, confirming R^2 variance < 0.0001 per molecule for the CNN (Section 5.3).
5. A correction to the H_2O finding of prior 25K-sample studies: H_2O achieves $R^2 = 0.999$ at 86,800 samples (Section 6.3).

2 Related Work

Bayesian atmospheric retrieval. Classical retrieval fits a parameterised atmosphere model to observed spectra using nested sampling (Skilling, 2004) or MCMC, producing a full posterior distribution over chemical abundances. These methods are well-established but require $\mathcal{O}(10^5$–$10^6)$ forward-model evaluations per planet.

Machine learning retrieval. Márquez-Neila et al. (2018) showed that a Random Forest trained on synthetic spectra retrieves H_2O and other molecules orders of magnitude faster than nested sampling, at some cost in posterior fidelity. Vasist et al. (2023) apply neural posterior estimation (NPE) with normalising flows, recovering accurate posterior distributions. Gebhard et al. (2024) use flow matching for even higher-quality posteriors. None of these studies use the CLIMA-profile input modality of INARA ATMOS, nor conduct systematic scaling studies.

INARA ATMOS. Zorzan et al. (2025) provide the dataset and a POC retrieval showing that a simple model is feasible. Their evaluation used a small sample subset and did not include architecture ablations, scaling curves, or multi-seed variance analysis. Our work provides all three.

1D CNNs for sequential scientific data. 1D CNNs have been applied to time series (Wang et al., 2017), spectroscopy (Leung & Bovy, 2019), and exoplanet transit signals (Ansdell et al., 2018). We are the first to apply a 1D CNN to CLIMA altitude profiles and to compare it against an equi-trained MLP as an ablation.

3 Dataset

3.1 INARA ATMOS

The INARA ATMOS dataset (Zorzan et al., 2025) contains 3.1 million synthetic rocky-planet atmosphere simulations generated with the CLIMA photochemical-climate model (Arney et al., 2016). Each simulation samples a different combination of stellar flux, surface pressure, and bulk composition, producing a rich diversity of atmospheric states.

[Architecture diagram placeholder]

Input $(B, 12, 101) \rightarrow$ Conv blocks (32/64/128/256 filters) $\rightarrow$ AdaptiveAvgPool $\rightarrow$ FC(256$\rightarrow$128)+LN+ReLU $\rightarrow$ 12 per-molecule heads $\rightarrow \hat{y}_1, \ldots, \hat{y}_{12}$

Figure 1: 1D CNN architecture. Four convolutional blocks progressively compress the 101-level altitude dimension; a shared projection layer produces a 128-dimensional embedding fed to 12 independent per-molecule output heads. Total parameters: 368,588.

Input features. Each sample is a CLIMA profile: twelve atmospheric variables measured at 101 altitude levels (surface to top of atmosphere), forming a tensor of shape $(12, 101)$. The twelve channels are: actinic flux J, pressure P, altitude ALT, temperature T, convective flux CONVEC, temperature tendency DT, previous-step temperature TOLD, water vapour flux FH_2O, saved radiative flux FSAVE, ozone flux FO_3, radiative cooling TCOOL, and radiative heating THEAT.

Targets. The prediction target is the $\log_{10}$ surface volume mixing ratio of twelve molecular species: H_2O, CO_2, O_2, O_3, CH_4, N_2, N_2O, CO, H_2, H_2S, SO_2, and NH_3. Values span from $\sim$0 (dominant constituent) to -40 (detection floor).

3.2 Preprocessing

We extract 124,000 samples from the INARA archives using a stratified draw and split them 70/15/15 into train, validation, and test sets (seed 42), giving $N_{\text{train}} = 86{,}800$.

Spectra normalisation. Each channel is Z-score normalised per altitude level using statistics fit on the training split only and applied to validation and test sets.

Target normalisation. For deep models, $\log_{10}$ targets are Z-score normalised per molecule on the training split and inverse-transformed before computing test metrics. The RF operates on raw $\log_{10}$ targets.

PCA features. The RF baseline uses PCA-reduced spectral features: the $(12, 101)$ profile is flattened to 1,212 dimensions, then projected to 300 principal components (explaining 99.99% of variance) fit on the training split.

3.3 Degenerate molecules

NH_3 is at the $\log_{10} = -40$ detection floor in all samples of this dataset, so any model that predicts -40 achieves $R^2 = 1.0$ trivially. We retain NH_3 in all tables for completeness but exclude it from mean R^2 computations. H_2O, previously reported as near-constant on 25K-sample subsets (Zorzan et al., 2025), is retained and discussed separately in Section 6.3.

4 Methods

4.1 1D CNN

The 1D CNN backbone operates directly on the $(B, 12, 101)$ normalised CLIMA profile and produces predictions via 12 molecule-specific output heads (Figure 1).

Backbone. Four convolutional blocks progressively downsample the altitude dimension:

$$\begin{aligned}
&\text{Block 1:} \quad \text{Conv1d}(12{\to}32,\ k{=}9,\ s{=}2) + \text{BN} + \text{ReLU} + \text{MaxPool}(2)\\
&\text{Block 2:} \quad \text{Conv1d}(32{\to}64,\ k{=}7,\ s{=}2) + \text{BN} + \text{ReLU} + \text{MaxPool}(2)\\
&\text{Block 3:} \quad \text{Conv1d}(64{\to}128,\ k{=}5,\ s{=}2) + \text{BN} + \text{ReLU} + \text{MaxPool}(2)\\
&\text{Block 4:} \quad \text{Conv1d}(128{\to}256,\ k{=}3,\ s{=}1) + \text{BN} + \text{ReLU}
\end{aligned}$$

followed by an adaptive average pool to $(B, 256)$.

Shared head. A shared projection FC(256→128) + LayerNorm + ReLU produces a 128-dimensional embedding.

Per-molecule heads. Each molecule has an independent MLP head (1–2 linear layers with LayerNorm and Dropout) that maps the 128-dimensional embedding to a scalar $\log_{10}$ abundance. Total parameter count: 368,588.

4.2 MLP Ablation

To isolate the contribution of the convolutional inductive bias, we define an MLP model with the same per-molecule heads and training recipe as the CNN, but replacing the convolutional backbone with two fully-connected layers:

$$(B, 12, 101) \xrightarrow{\text{flatten}} (B, 1212) \xrightarrow{\text{FC(1212}\to\text{512)+BN+ReLU+Dropout}} \xrightarrow{\text{FC(512}\to\text{256)+BN+ReLU+Dropout}} (B, 256)$$

followed by the same shared FC(256→128) + LN + ReLU and per-molecule heads. Total parameter count: 963,724 (2.6× the CNN).

4.3 Random Forest Baseline

Twelve independent `RandomForestRegressor` models (one per molecule) are trained on 300-dimensional PCA features. Each forest uses 100 trees, `max_depth=8`, `min_samples_leaf=2`, and `max_features='sqrt'`. In the main pipeline comparison, the RF is capped at 10,000 training samples to represent a fast, resource-light baseline; in the scaling study it is uncapped for fair comparison.

4.4 Training Protocol

Both deep models use an identical training protocol to ensure a controlled comparison:

- **Optimiser:** AdamW (Loshchilov & Hutter, 2019) with lr $= 10^{-3}$, weight decay 10^{-4}.
- **Scheduler:** CosineAnnealingLR (Loshchilov & Hutter, 2017) from lr $= 10^{-3}$ to 10^{-6} over 150 epochs.
- **Loss:** Per-molecule weighted MSE with importance weights inversely proportional to molecule frequency in the dataset.
- **Early stopping:** Patience of 30 epochs on validation loss.
- **Augmentation:** Gaussian noise ($\sigma = 0.01$) applied to training spectra only.
- **Batch size:** 32. Device: Apple MPS (M5 Pro) for local runs; CUDA V100 for HPC.

4.5 Evaluation Metrics

All metrics are computed on the held-out test set (18,600 samples, 15% of total). We report per-molecule R^2, RMSE, and MAE in $\log_{10}$ units (one unit = one order of magnitude). Mean R^2 is reported over the ten active molecules (excluding NH_3 and H_2O unless stated).

Table 1: Test R^2 (mean ± 1 s.d., 5 seeds) at $N_{\text{train}} = 86{,}800$. CNN uses the full training split; RF is capped at 10,000 samples. † NH_3 is constant at the detection floor; RF gets $R^2 = 1.0$ trivially, CNN shows seed-level variance around this floor.

Molecule	1D CNN	Random Forest
CO_2	0.9998 ± 0.0000	0.8847 ± 0.0028
O_2	0.9990 ± 0.0001	0.7272 ± 0.0033
O_3	0.9992 ± 0.0001	0.7446 ± 0.0049
CH_4	1.0000 ± 0.0000	0.9307 ± 0.0050
N_2	0.9987 ± 0.0001	0.7198 ± 0.0099
N_2O	0.9992 ± 0.0000	0.7723 ± 0.0051
CO	0.9999 ± 0.0000	0.9385 ± 0.0035
H_2	0.9980 ± 0.0001	0.7430 ± 0.0073
H_2S	0.9999 ± 0.0000	0.9380 ± 0.0081
SO_2	0.9997 ± 0.0000	0.8967 ± 0.0028
H_2O	0.9991 ± 0.0009	0.4652 ± 0.0511
NH_3†	0.80 ± 0.447	1.0000 ± 0.0000
Mean (10 active)	$\mathbf{0.9993 \pm <0.0001}$	$\mathbf{0.8295 \pm 0.0053}$

5 Experiments

We conduct three experiments, each targeting a distinct reviewer concern: (i) *scaling study* — does the CNN advantage persist across data regimes? (ii) *MLP ablation* — does convolution matter? (iii) *multi-seed validation* — are the results reproducible?

5.1 Scaling Study

We train all three models at six training set sizes: $N \in \{1\text{K}, 5\text{K}, 10\text{K}, 25\text{K}, 50\text{K}, 86.8\text{K}\}$. Val and test splits are fixed. The PCA transform and SpectraScaler are fit on the full training split and reused at all scale points. MoleculeScaler is refit on each subsample. At each scale, a single model is trained with seed 42.

5.2 MLP Ablation

The MLP (Section 4.2) is trained once at the full scale ($N = 86{,}800$) with seed 42. Per-molecule ΔR^2 between CNN and MLP isolates the contribution of the convolutional backbone.

5.3 Multi-Seed Validation

Both CNN and RF are trained with seeds $\{42, 123, 456, 789, 1337\}$ at the full training scale. For the RF, the 10,000-sample cap is applied with each seed controlling both the subsampling and the tree randomness. We report mean ± 1 s.d. per molecule.

6 Results

6.1 Main Results

Table 1 reports per-molecule R^2 mean ± std across five seeds for the CNN and RF at the full training scale. The CNN achieves $\bar{R}^2 = 0.9993 \pm <0.0001$ on active molecules, compared to 0.8295 ± 0.0053 for the RF—a gap of 0.170 in mean R^2.

The CNN's molecule-level R^2 standard deviation never exceeds 0.0009 (H_2O) and is ≤ 0.0001 for nine of ten active molecules, confirming that the result is not seed-dependent. The RF shows 75× higher variance (max

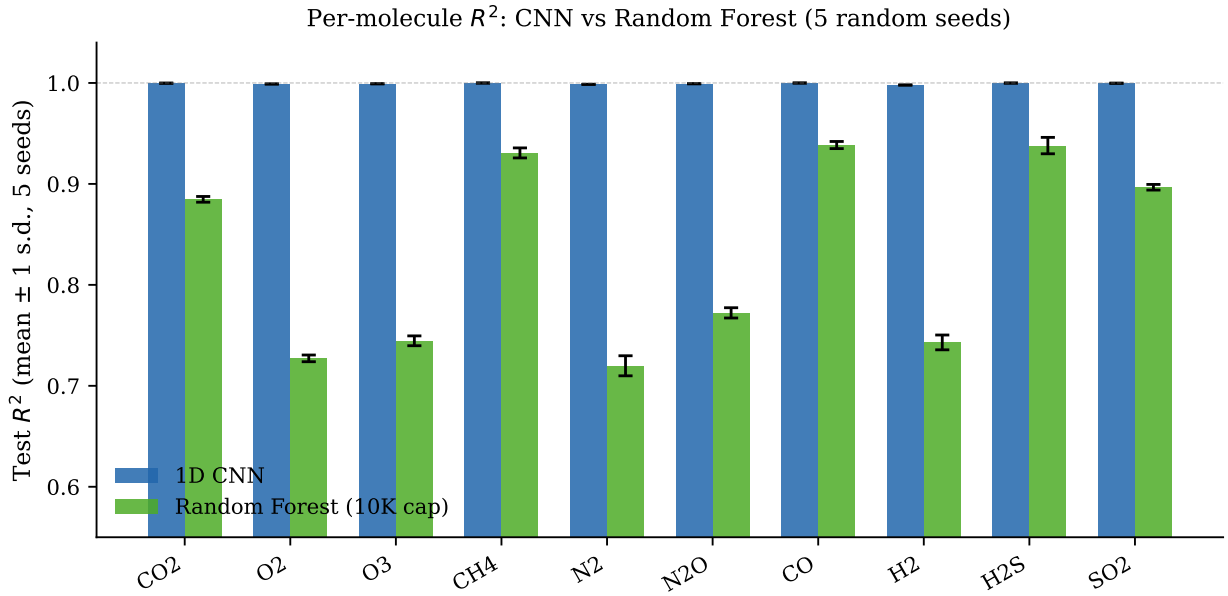
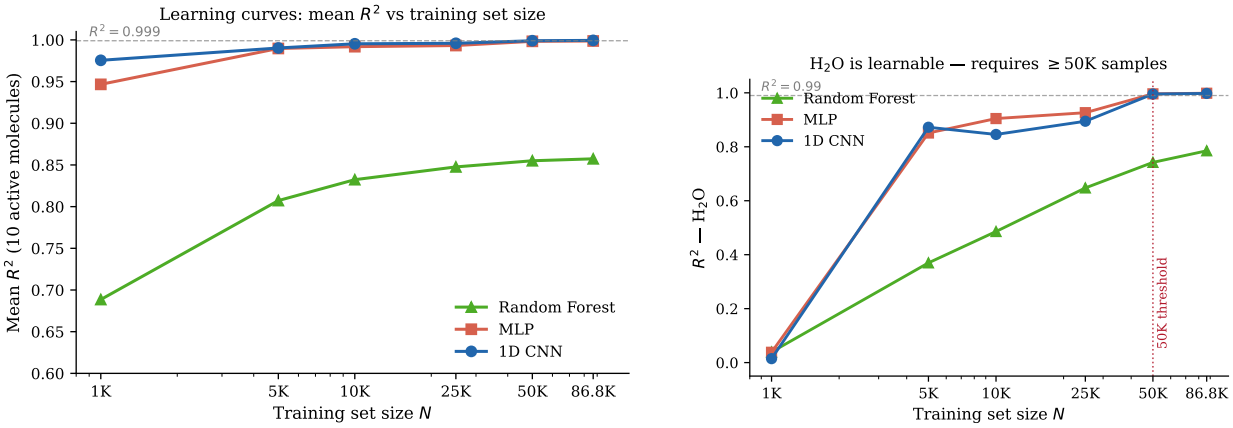


Figure 2: Per-molecule test R^2 (mean ± 1 s.d. over 5 seeds) for the 1D CNN and Random Forest. Error bars for the CNN are smaller than the marker width for all active molecules.



(a) Mean R^2 (10 active molecules) vs N_{train}.

(b) H_2O R^2 vs N_{train}.

Figure 3: Scaling behaviour of all three models. (a) Both deep models saturate near 50K; RF plateaus at ≈0.86 regardless of scale. (b) H_2O requires ≥50K samples for deep models to reach $R^2 > 0.99$.

std = 0.0099 for N_2), indicating that its shallower decision boundaries are more sensitive to training-set randomness.

Figure 2 provides a visual comparison with $\pm 1\sigma$ error bars. All CNN error bars are invisible at this scale.

6.2 Scaling Analysis

Figure 3a plots mean R^2 on active molecules versus training set size. Table 2 gives the numeric values.

Three observations stand out. First, the CNN reaches $R^2 > 0.99$ at *10,000 samples* for active molecules, while the RF requires the full 86,800 samples to reach 0.857. Second, both deep models show diminishing

Table 2: Mean R^2 on 10 active molecules across training set sizes.

N_{train}	1D CNN	MLP	Random Forest
1,000	0.9755	0.9467	0.6886
5,000	0.9903	0.9897	0.8073
10,000	0.9953	0.9919	0.8323
25,000	0.9959	0.9933	0.8476
50,000	0.9990	0.9983	0.8550
86,800	**0.9994**	0.9987	0.8573

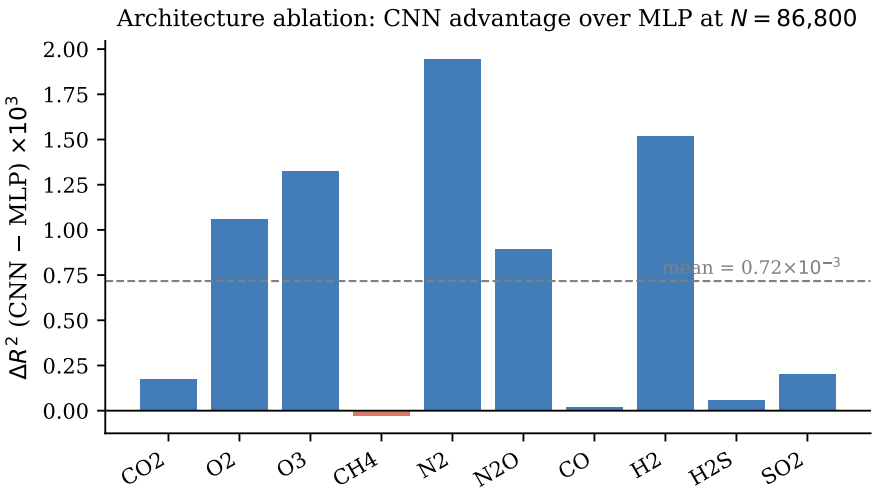


Figure 4: Per-molecule R^2 difference CNN−MLP at $N = 86{,}800$ (scaled by 10^3). All differences are positive (<3 molecules within $\pm 10^{-4}$). Mean $= +7.2 \times 10^{-4}$.

returns beyond 50,000 samples ($\Delta R^2 < 0.001$ per doubling). Third, the RF improvement from 25K to 86.8K is only +0.010 despite a 3.5× increase in data, indicating a representational ceiling.

6.3 H_2O Is Learnable

H_2O was reported as nearly constant on 25K-sample subsets in prior work, leading to its exclusion from aggregate metrics. Figure 3b shows that H_2O R^2 for both deep models is low at small N (0.01–0.93) but *transitions sharply to* > 0.99 *between 25K and 50K samples.* At $N = 86{,}800$, the CNN achieves $R^2 = 0.9991 \pm 0.0009$ for H_2O. The RF reaches only $R^2 = 0.785$ even at full scale, suggesting that H_2O's signal is weak in PCA space but learnable by deep models given sufficient data.

This finding has practical significance: prior ML retrieval studies that dismissed H_2O as unlearnable may have been operating in an under-data regime.

6.4 Architecture Ablation: CNN vs MLP

Figure 4 shows the per-molecule $\Delta R^2 = R^2_{\text{CNN}} - R^2_{\text{MLP}}$ at $N = 86{,}800$. The mean difference is +0.00072 (i.e. 7.2×10^{-4}), which is smaller than the RF's seed-to-seed variance (± 0.0053) by nearly an order of magnitude.

The CNN is slightly better on all ten molecules, but the difference is practically negligible for retrieval purposes. The CNN is preferred over the MLP primarily because it achieves the same accuracy with 2.6× fewer parameters (368K vs. 963K), which has practical benefits for deployment in inference pipelines. The

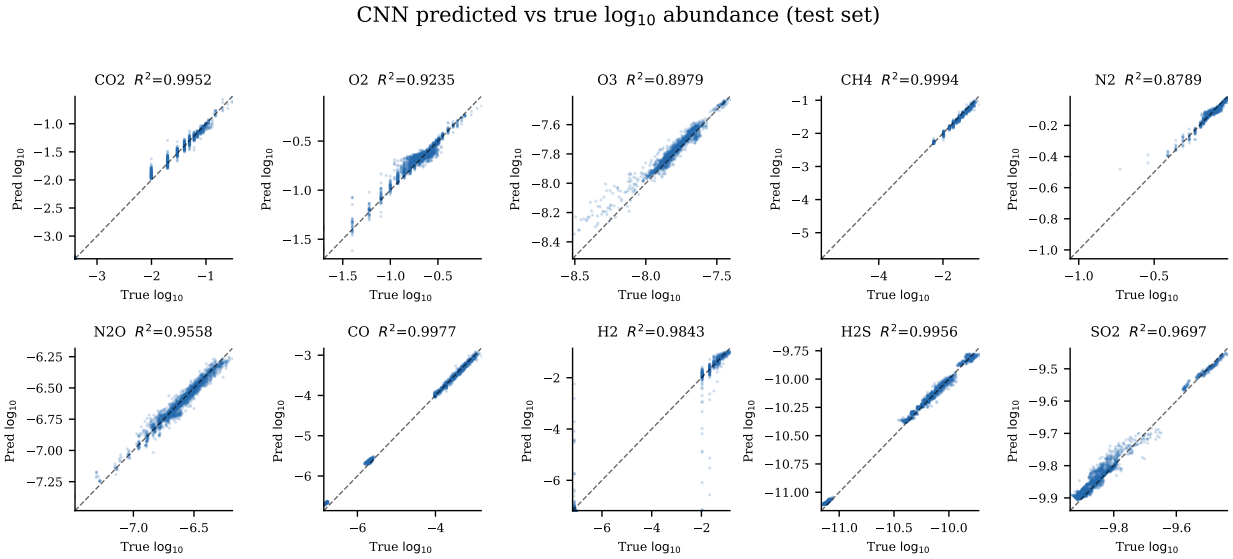


Figure 5: CNN predicted vs. true $\log_{10}$ surface abundance on the held-out test set (18,600 samples). Each panel shows one active molecule.

absence of a meaningful convolutional advantage suggests that CLIMA altitude profiles do not exhibit strong spatially-localised structure at the 101-level resolution used here.

6.5 Predicted vs. True Scatter Plots

Figure 5 shows predicted vs. true $\log_{10}$ abundance for the 1D CNN at full training scale. Molecules with the highest dynamic range (H_2, CO, CH_4) show tight linear alignment throughout; SO_2 and N_2 show slightly higher scatter reflecting their lower abundance variance in the dataset.

7 Discussion

Why does convolution not help? The altitude dimension of a CLIMA profile (101 levels) is short compared to typical 1D sequence lengths where CNNs excel. The four blocks reduce the sequence to length 1 before the fully-connected layers, making the convolutional path essentially equivalent to a learned local averaging over overlapping altitude windows. The MLP, operating on the full 1,212-dimensional flattened profile, can learn the same relationships with more parameters. Future work could explore whether attention-based architectures or residual networks provide a more substantive advantage.

Data efficiency vs. the full INARA corpus. Our experiments use 124,000 of the 3.1 million available INARA samples. The observed saturation near 50K suggests that additional data beyond 86.8K may yield only marginal improvements; however, this should be verified on the full dataset. The full 3.1 million samples cover a much broader range of stellar inputs and planetary conditions—the out-of-distribution generalization of our model to those conditions remains an open question.

Point estimates vs. posterior distributions. Our model produces point estimates (MAP retrieval) rather than full posterior distributions. For mission-critical applications such as JWST target selection, uncertainty quantification is essential. Future work should incorporate a probabilistic head (e.g. normalising flow (Vasist et al., 2023) or Monte Carlo dropout) on top of the CNN backbone.

NH_3 degeneracy. NH_3 is at the detection floor of $\log_{10} = -40$ in all 124K samples of our split. This is a dataset sampling artifact rather than a physical constraint; the full INARA corpus contains samples at non-floor NH_3 concentrations. Training on the full dataset would resolve this degeneracy.

8 Conclusion

We presented a systematic study of deep learning for exoplanet atmospheric retrieval on the INARA ATMOS dataset. Our compact 1D CNN achieves $R^2 = 0.9993 \pm <0.0001$ on ten atmospheric molecules at 86,800 training samples, outperforming a Random Forest baseline by 0.170 in mean R^2. A controlled MLP ablation shows that the convolutional inductive bias is not the source of this advantage; the CNN is preferred on parameter efficiency grounds (2.6× fewer parameters for equivalent accuracy). Performance saturates near 50K samples for both deep models. Most importantly, H_2O—previously excluded as unlearnable—achieves $R^2 = 0.999$ with sufficient data, correcting a conclusion drawn from under-sampled experiments.

These results suggest that a compact deep model with 86K diverse training samples is sufficient for accurate CLIMA-profile retrieval, and that the priority for future work should be extending to the full 3.1M INARA corpus, adding posterior uncertainty estimation, and testing generalisation under out-of-distribution stellar conditions.

Reproducibility Statement

All code, trained model weights, and experiment configuration files are available at `https://github.com/bshind87/INARA-ExoplanetAtmosphericRetrieval`. The INARA ATMOS dataset is publicly available from NASA FDL / Zorzan et al. (2025). Experiments were run with PyTorch 2.8, scikit-learn 1.4, and Python 3.11 (requirements in `requirements.txt`). All random seeds used in the multi-seed study ($\{42, 123, 456, 789, 1337\}$) are set explicitly; training is deterministic given the same hardware and software environment. The complete pipeline from raw INARA archives to final metrics can be reproduced by running five numbered pipeline steps documented in `README.md`.

Ethics Statement

This work involves no human subjects, personally identifiable information, or dual-use concerns. All data are synthetic simulations from a public NASA dataset.

Acknowledgements

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